

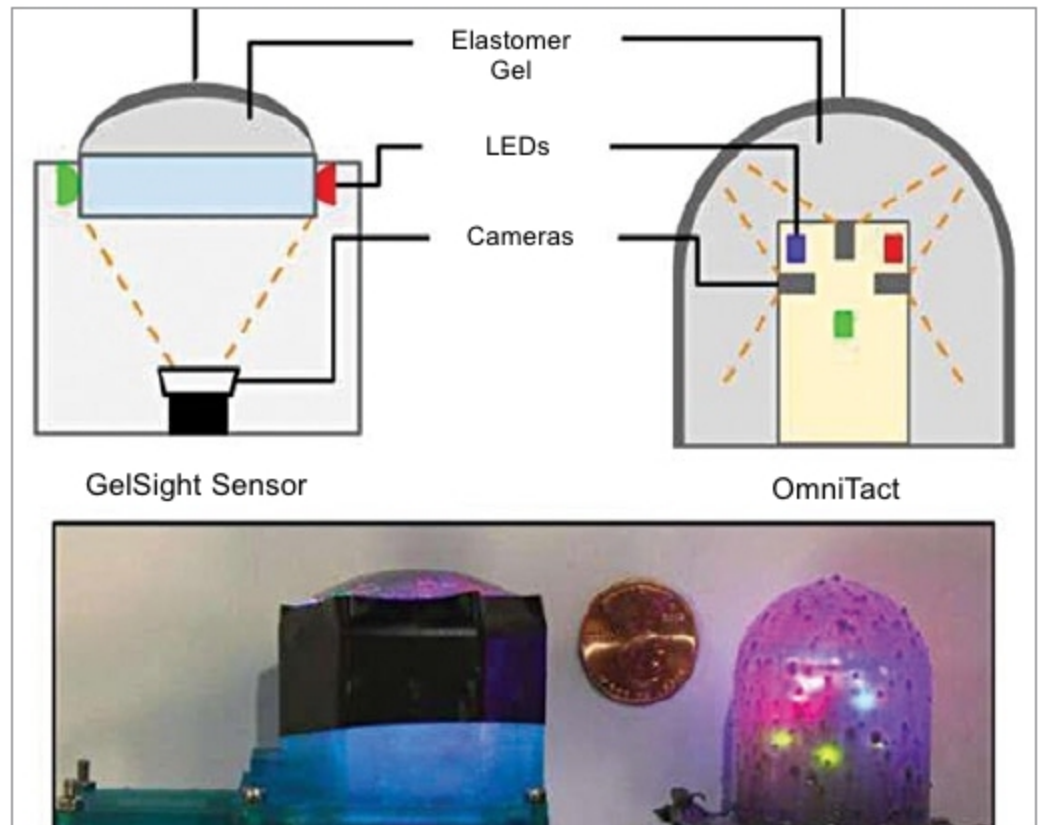
## OmniTact: A high-resolution tactile sensor for robotics applications

From the past few years, researchers worldwide have been trying to develop sensors that could replicate humans' sense of touch in robots and enhance their manipulation skills. While some of these sensors achieved remarkable results, most existing solutions have small sensitive fields or can only gather images with low resolutions.

A team of researchers at UC Berkeley recently developed a new multi-directional tactile sensor, called OmniTact, that overcomes some of the limitations of previously developed sensors. OmniTact, presented in a paper pre-published on arXiv and set to be presented at ICRA 2020, acts as an artificial fingertip that allows robots to sense the properties of objects it is holding or manipulating. "Our lab recognised the need for a sensor for general robotic manipulation tasks with expanded capabilities than current tactile sensors," Frederik Ebert, one of the researchers who carried out the study, told TechXplore.

OmniTact, the sensor developed by Ebert and his colleagues, is an adaptation of GelSight, a tactile sensor created by researchers at MIT and UC Berkeley. GelSight can generate detailed 3D maps of an object's surface and detect some of its characteristics.

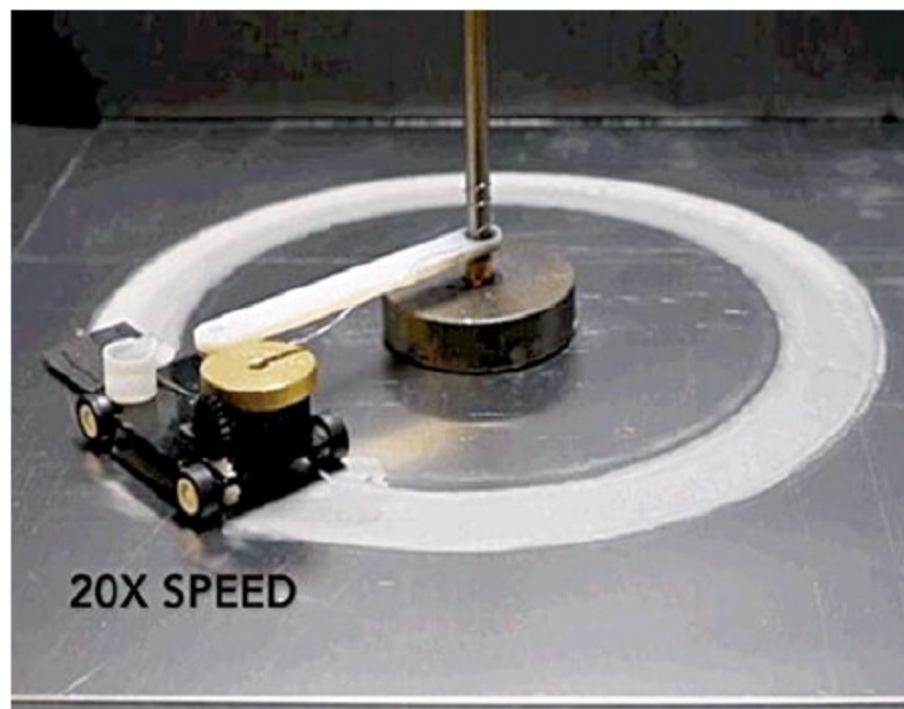
In contrast with GelSight, OmniTact is multi-directional, which means that all of its sides have sensing capabilities. In addition, it can provide high-resolution readings, is highly



An illustration and image explaining the basic differences between the GelSight Sensor and OmniTact (Credit: Padmanabha et al)

compact and has a curve shape. When integrated into a gripper or robotic hand, the sensor acts as a sensitive artificial 'finger,' allowing the robot to manipulate and sense a wide range of objects varying in shape and sizes.

## New scavenger technology allows robots to eat metal for energy



Rather than a battery, the researchers' metal-air scavenger vehicle gets energy from breaking chemical bonds in the aluminium surface it travels over (Credit: University of Pennsylvania)

When electronics need their own power sources, there are two basic options: batteries and harvesters. Batteries store energy internally, but are therefore heavy and have a limited supply. Harvesters, such as solar panels, collect energy from their environments. This gets around some of the downsides of batteries but introduces new ones, in that they can only operate in certain conditions and can't turn that energy into useful power very quickly.

New research from the School of Engineering and Applied Science is bridging the gap between these two fundamental technologies for the first time in the form of a 'metal-air scavenger' that gets the best of both worlds.

This metal-air scavenger works like a battery, in that it provides power by repeatedly breaking and forming a series of chemical bonds. But it also works like a harvester, in that power is supplied by energy in its environment, specifically, the chemical bonds in metal and air surrounding the metal-air scavenger. The result is a power source that has ten times more power density than the best energy harvesters and thirteen times more energy density than lithium-ion batteries.